



OptiBench: An Optimization Benchmark Tool for Renewable Energy Problems

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Background

- Benchmarking the computational performance of software algorithms and hardware devices is an interdisciplinary practice.
- Advancements in optimization solvers, including the use of GPUs, have not been well quantified across a diverse set of problems.
- Current benchmarking solutions need to be adapted to the diverse range of renewable energy problems to study the efficient use of compute capabilities including accelerators.



Motivation

Photo by Joe DelNero / NREL

Develop a user-friendly benchmarking tool to evaluate performance of optimization solvers in solving renewable energy problems on an HPC.

Objectives

- Develop a tool to benchmark renewable energy problems across a wide range of optimization solvers.
- Post-process and visualize benchmarking results for easy analysis.
- Utilize High-Performance Computing (HPC) resources to accelerate benchmarking speed and expand capacity.
- Support GPU-based optimization solvers.

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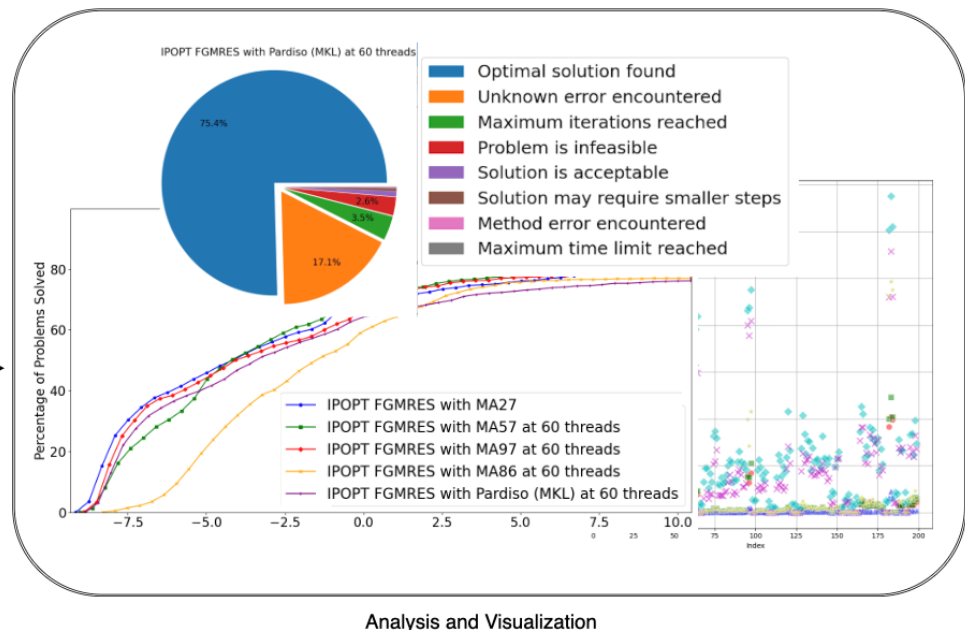
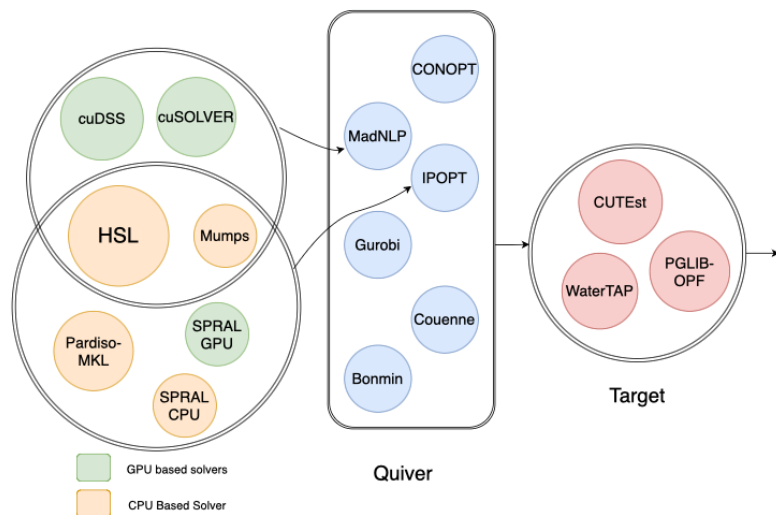
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OptiBench

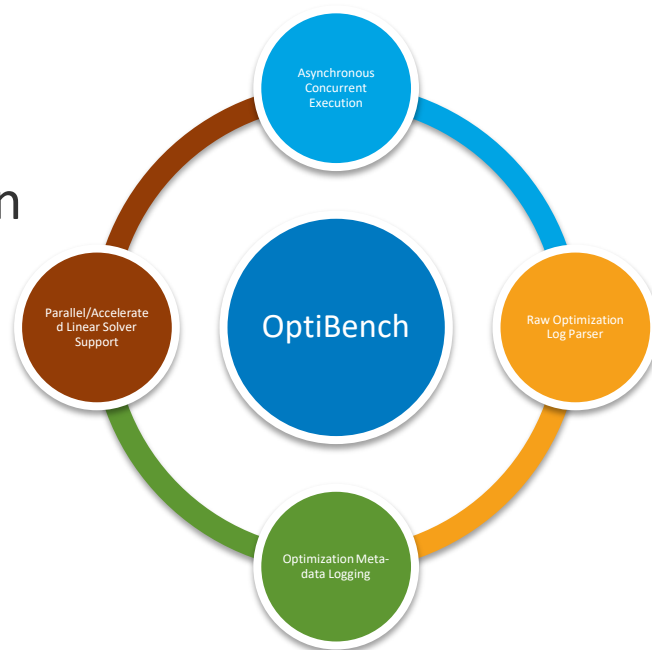
ReFrame Test



Core features

OptiBench offers following core features:

1. Asynchronous concurrent execution
2. Raw optimization log parser
3. Buffered meta-data logging
4. Parallel/accelerated solver testing



Quiver: set of optimization solvers

- OptiBench is compatible with any optimization solvers supported by the JuMP or NLPModels frameworks.
- OptiBench can directly benchmark with optimization solver executables, in case the optimization solver is not yet supported by Julia (such as Uno).

Tested solver	Linear solvers
IPOPT	HSL, Mumps, Pardiso (MKL), SPRAL
Bonmin	HSL, Mumps
Couenne	HSL, Mumps
MadNLP	HSL, Mumps, SPRAL
Gurobi	N/A
KNITRO	HSL, Pardiso (MKL)
Xpress	N/A
CONOPT	N/A
Uno	MA57
SCIP	N/A
Alpine	Same as sub-optimizer

Target: set of benchmarks

- OptiBench can benchmark any model in JuMP or NLPModels framework and read any model in the file formats listed in the table to the right.
- OptiBench also supports any file format supported by optimization solver executable directly.

File format	File extension
Conic benchmark format	.cbf
Linear programming format	.lp
MathOPT format	.mof.json
AMPL NL	.nl
REW	.rew
SDPA	.sdpa
SIF	.sif
Power Models file format	.m

Benchmarks

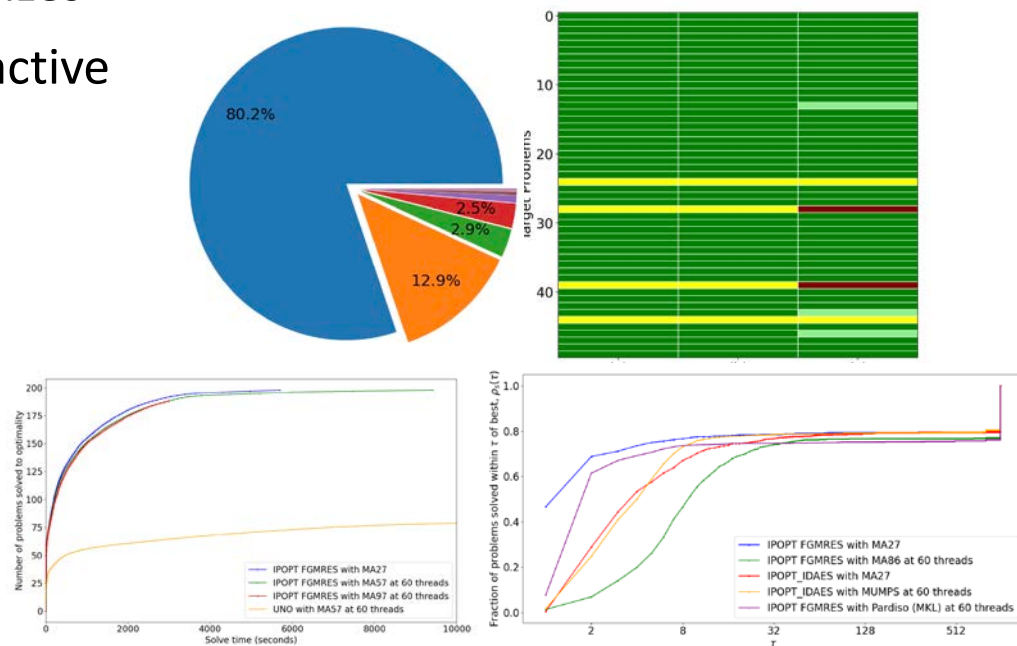
OptiBench has been evaluated with three benchmark suites of interest to renewable energy research.

- **CUTEst:** Constrained and unconstrained testing environment with safe threads
- **PGlib-OPF:** Power grid library for benchmarking optimal power flow algorithms.
- **WaterTAP:** Open-source water treatment model library

Post-processor

OptiBench's post-processor visualizes benchmarking results using interactive and intuitive plots.

- Cumulative time plots
- Performance profiles
- Robustness pie plots
- Heat maps
- Scatter plots



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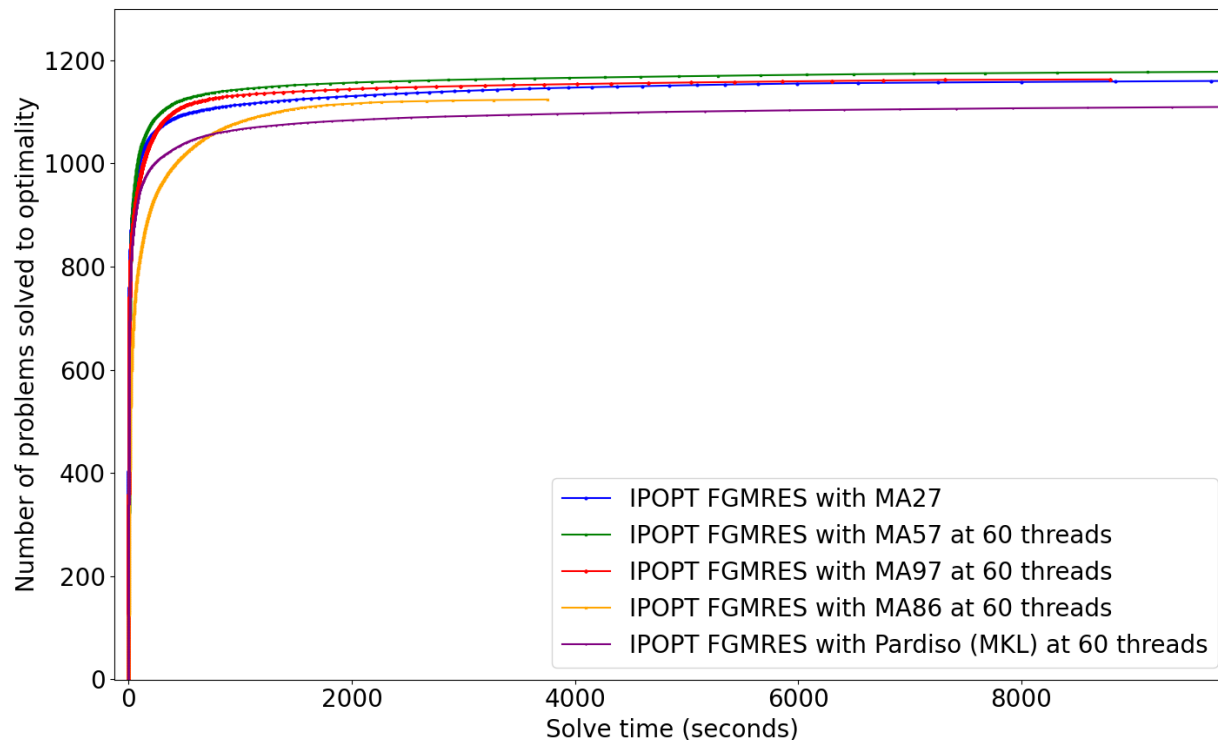
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Demonstration

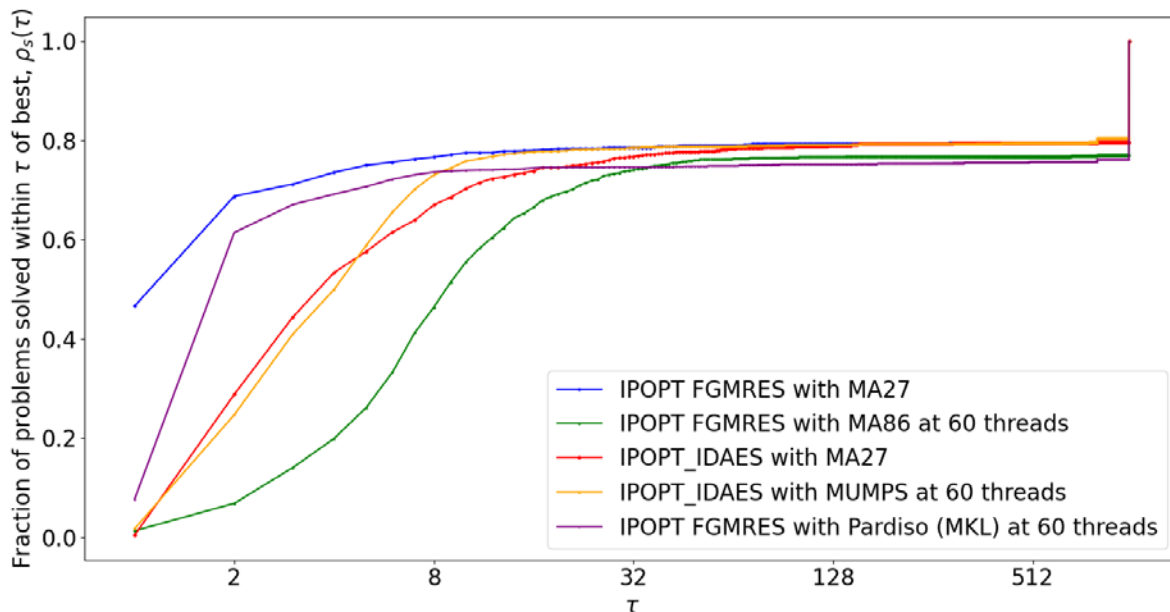
- **NREL's Kestrel HPC CPU nodes:** HPE Cray EX system, 2×52 dual socket Intel Xeon Sapphire Rapids processor and 256 GB RAM.
- 1 node with 104 tasks per node and Julia v1.10.0 was used for the benchmark run.
- Benchmark sizes:
 - CUTEst: 1531 problems
 - PGLib-OPF: 199 problems
 - WaterTAP: 72 problems
- A total of 10 solver-linear solver combinations were compared in this section.

Cumulative time plots: CUTEst



Experimental IPOPT
FGMRES with MA57
yields the best
performance.

Performance profiles: CUTEst



Performance ratio, $\rho_s(\tau)$ for a given optimization solver s is defined as:

$$r_{p,s} = \begin{cases} \frac{t_{p,s}}{\min\{t_{p,s} : s \in \mathcal{S}\}} & \text{if convergence test passed} \\ \infty & \text{if convergence test failed} \end{cases}$$

Which then defines performance profile as:

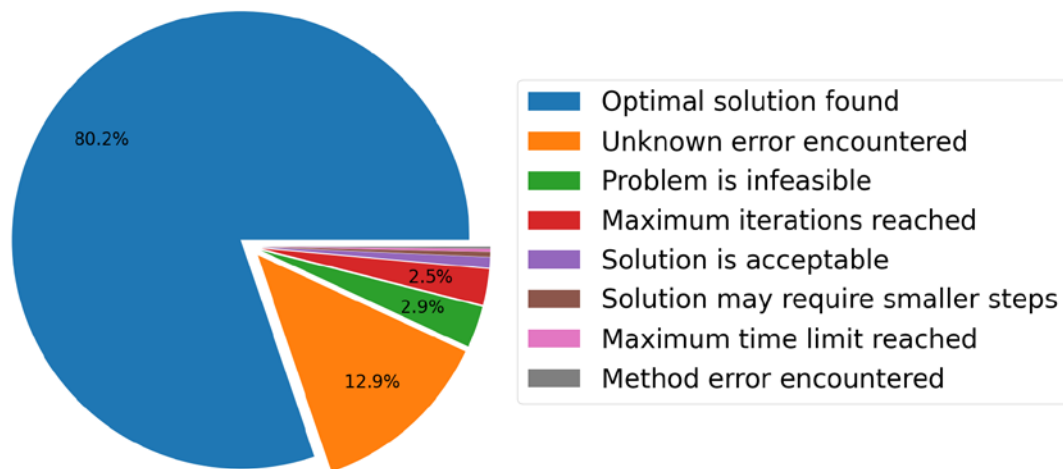
$$\rho_s(\tau) = \frac{|\{p \in \mathcal{P} : r_{p,s} \leq \tau\}|}{|\mathcal{P}|},$$

IPOPT FGMRES with MA27 yields the best performance.

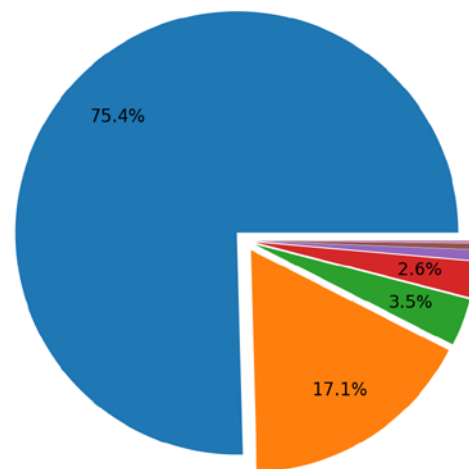
Robustness pie plots: CUTEst

Best vs. worst performer For CUTEst:

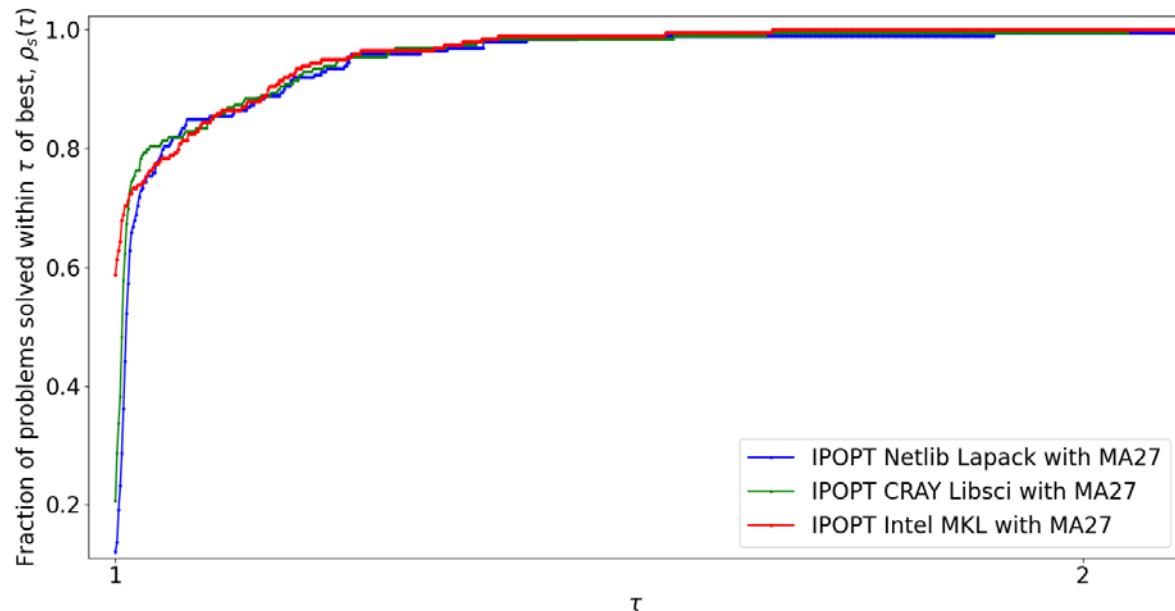
(a) IPOPT FGMRES with MA57 at 60 threads



(b) IPOPT FGMRES with Pardiso (MKL) at 60 threads



Performance profiles: PGLib-OPF

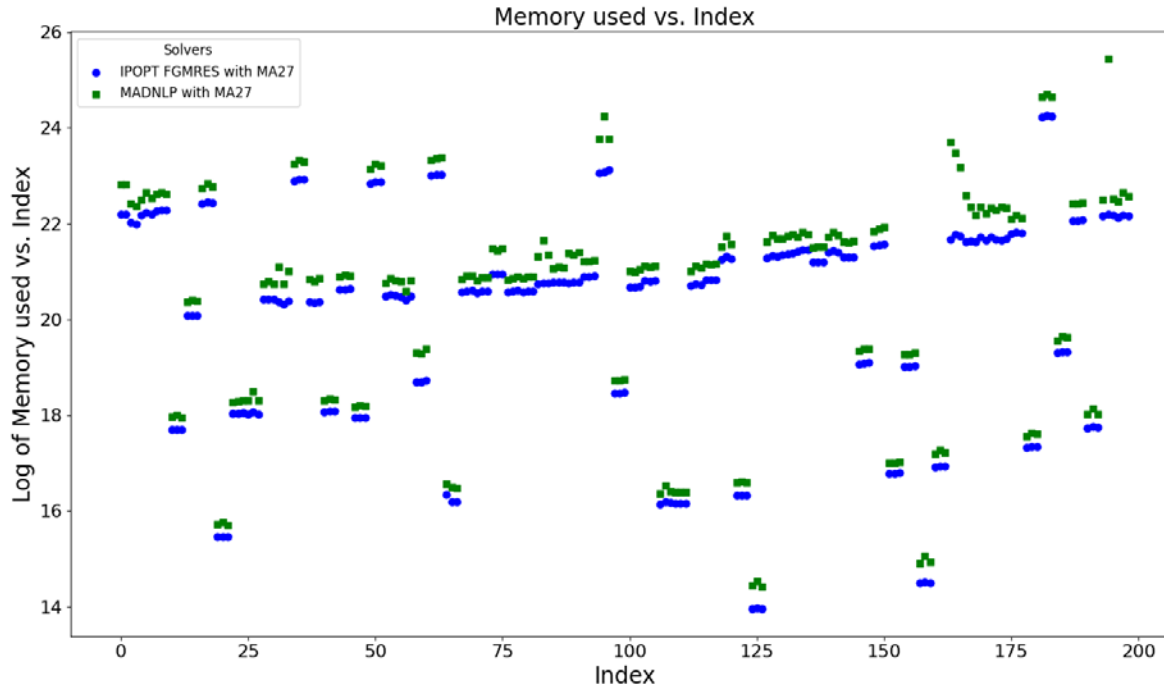


All three IPOPT v3.13.2 variations seems to be taking top spot for certain values of τ .

Total solve time statistics: PGLib-OPF

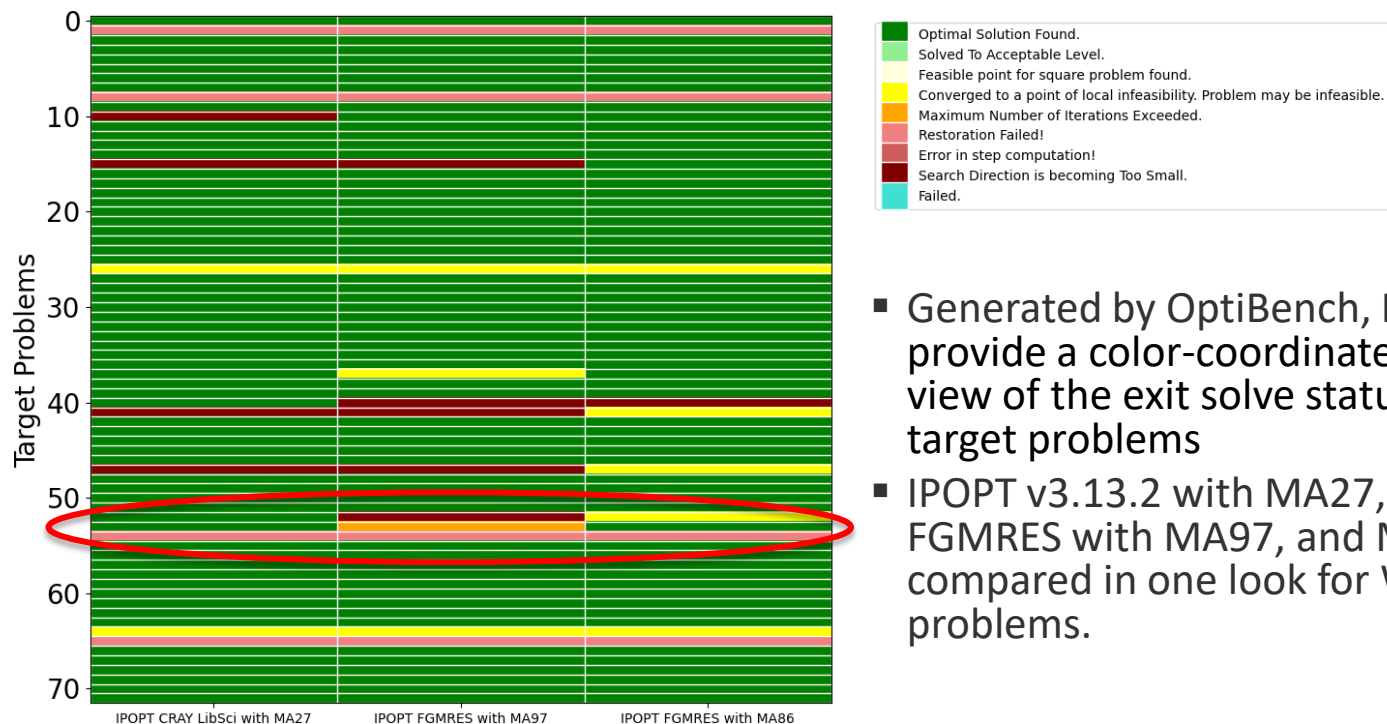
Optimizer	MA27		MA57		Mumps	
	Time (in sec)	Performance Ratio	Time (in sec)	Performance ratio	Time (in sec)	Performance ratio
ILOPT CRAY Libsci	7429.94	1	9939.68	1.337	11098.76	1.493
ILOPT Intel MKL	7429.53	1	10074.08	1.355	11266.59	1.516
ILOPT Netlib Lapack	7468.85	1.005	10077.08	1.356	11319.61	1.523

Memory usage: PGLib-OPF



IPOPT FGMRES uses lesser memory than MadNLP with the same linear solver MA27.

Heat maps: WaterTAP



- Generated by OptiBench, Heatmaps provide a color-coordinated tabulated view of the exit solve statuses of the target problems
- IPOPT v3.13.2 with MA27, and IPOPT FGMRES with MA97, and MA86 can be compared in one look for WaterTAP problems.

Demonstration results

- Based on the comparison following conclusions can be made:
 - For CUTEst, IPOPT FGMRES performs the best with MA57.
 - Out of the three mathematical libraries CRAY LibSci, and Intel MKL performed superior to the general purpose Netlib LAPACK library on Kestrel CPU nodes.
 - For PGLib-OPF with IPOPT v3.13.2, linear solver MA27 performs the fastest followed by MA57 and Mumps, respectively.

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Conclusion

- OptiBench is capable of benchmarking well recognized benchmarks against state-of-the-art optimization solvers.
- The concurrent execution and linear solver support allows OptiBench to maximize the utilization of computational resources.
- It can post process and visualize the results with minimal user intervention.
- OptiBench can be integrated with ReFrame regression test allowing it to leverage of ReFrame on HPC.

Way forward

- Expand support to a wider range of optimization algorithms, benchmarks, and introduce diverse methods for result analysis.
- Explore the potential of using the tool for parameter tuning or a diagnostic toolbox for nonlinear problems and optimization algorithms.

Thank you

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